

Bowen Xie

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RESEARCH INTERESTS

Applied Probability, Queueing Theory, Stochastic Control, Operations Research, Mathematical Finance, Reinforcement Learning, Numerical Analysis

EDUCATION

Doctor of Philosophy	Iowa State University , Department of Mathematics	Aug 2017 -Aug 2022
	- Applied Mathematics, GPA: 3.92/4.00 - Advisor: Dr. Ananda Weerasinghe 🔗	
Master of Science	Iowa State University , Department of Mathematics	Aug 2017 -May 2019
	- Applied Mathematics, GPA: 3.90/4.00 - Advisor: Dr. Ananda Weerasinghe 🔗	
Bachelor of Science	Liaoning Technical University , Department of Mathematics	Sep 2013 -July 2017
	Mathematics & Applied Mathematics, GPA: 3.51/4.00 (Math GPA: 3.77/4.00)	

EXPERIENCE

The University of Akron , Assistant Professor	Akron, OH
Department of Mathematics, College of Engineering & Polymer Science	Fall 2024-present
Washington University in St. Louis , Postdoctoral Lecturer	St. Louis, MO
Department of Statistics and Data Science	Fall 2022-Spring 2024
Iowa State University , Graduate Teaching/Research Assistant	Ames, IA
Department of Mathematics	Fall 2017-Summer 2022

MENTORING AND STUDENTS

Niraj Jaishwal	Akron, OH
- Undergraduate student from Computer Science	Fall 2025-present
Samuel Adjei Baiden	Akron, OH
- Master student from Applied Mathematics	Fall 2024-Spring 2026
Benno Lutz	Akron, OH
- Undergraduate student honors project	Fall 2025-Spring 2026

PUBLICATIONS AND PREPRINTS

Publications

- [15] Y. Bai, Y. Gao, B. Xie, and S. Luo, "The symmetry method for fractional airy equation," *Mathematical Methods in the Applied Sciences*, vol. 49, no. 14, pp. 16 072–16 087, 2026.
- [14] B. Xie and Y. Gao, "Numerical analysis for fractional riccati differential equations based on finite difference method," *Journal of Nonlinear Modeling and Analysis* <https://journal.global-sci.org/jnma>, vol. 7, no. 1, pp. 189–208, 2025.
- [13] B. Xie, "Multi-component matching queues in heavy traffic," *Queueing Systems*, pp. 1–47, 2024, (Part of a Special Issue: Product Forms, Stochastic Matching, and Redundancy).
- [12] B. Xie and Y. Gao, "Diffusion approximation of a special bandwidth sharing model via infinitesimal generators with a lifting-projection method," *Operations Research Letters*, vol. 52, p. 107 063, 2024.

- [11] B. Xie and Y. Gao, “On the long-run average cost minimization problem of the stochastic production-inventory model,” *Journal of Industrial and Management Optimization*, vol. 20(5), pp. 1823–1844, 2024.
- [10] B. Xie and Y. Gao, “Population growth model with fractional operator and its numerical analysis,” *Journal of Mathematical Study*, 2024, (to be appeared).
- [9] Y. Gao and B. Xie, “Numerical analysis for fractional bratu type equation with explicit and implicit methods,” *Mathematical Methods in the Applied Sciences*, vol. 46, no. 17, pp. 18 447–18 457, 2023, (corresponding author).

Preprints and Working Papers

- [8] S. Adjei Baiden, B. Xie, and Y. Gao, *Optimal dynamic pricing under double-ended reflected stochastic market potential*, (under review at IISE Transactions).
- [7] B. Xie, J. Chávez-Casillas, and J. E. Figueroa-López, *Market making in limit order books*, (Working paper).
- [6] S. Ray, R. Wu, and B. Xie, *Regulated multi-component matching queues with abandonment: Scaling limits and large deviations*, (near completion), 2026.
- [5] B. Xie and Y. Gao, *Joint pricing and innovation control in regulated recycling-rate diffusion*, (submitted to IISE Transactions), 2026.
- [4] B. Xie, *An infinitesimal generator approach on weak convergence of regulated multi-class matching systems*, arXiv preprint arXiv:2507.09789 (submitted to Journal of Applied Probability, R&R), 2025.
- [3] B. Xie and H. Yin, *Admission control for a single server waiting time process in heavy traffic*, arXiv preprint arXiv:2212.05674 (to be submitted to Queueing Systems), 2025.

Theses

- [2] B. Xie, “Topics of queueing theory in heavy traffic (ph. d thesis),” <https://dr.lib.iastate.edu/items/be350d6c-e38a-4793-8e40-33445c2cdb5f/full>, Ph.D. dissertation, 2022.
- [1] B. Xie, “Long-run average cost minimization of a stochastic processing system (master creative component),” <https://dr.lib.iastate.edu/entities/publication/3f5cb727-8a92-46c2-8343-9252e1e7c78c>, M.S. thesis, 2019.

SELECTED RESEARCH PROJECTS ---

Dynamics of limit order books in high-frequency trading [7]

Fall 2023-present

Postdoc Mentor: Dr. José E. Figueroa-López, Washington University in St. Louis
Joint work with Dr. Jonathan Chávez Website, University of Rhode Island

- We aim to construct a matching system framework for LOBs and implement algorithms to address the optimal placement problem faced by market makers.
- Study the performance of reinforcement learning (RL) in solving market-making problems with inventory risk.
- Attempt to train market-making agent using RL with a real-world stock market constructed from historical Limit Order Book (LOB) data.
- Attempt to conduct Neural Networks to approximate state-value and state-action value functions and compare their performance with tile coding as well as the theoretical solution to the optimal placement problem.

Joint stochastic control on product and process innovations [5]

Spring 2025-present

- Production and process innovation are key drivers of quality improvement and cost reduction. In the circular economy, dynamic demand and resource recycling are crucial for operational efficiency.
- We explore stochastic control strategies within the propagation of chaos framework proposed in [5].
- The joint control of the product price and innovations under constraints such as recycling rates and demand propagation will be addressed.
- Our approach integrates rigorous stochastic analysis with the Actor-Critic

algorithm to develop adaptive and tractable solutions for complex systems.

- Applications in Monopoly's product and process innovations [8].

Scaling-limits on regulated matching systems [6]

Fall 2024-present

Joint work with Dr. Souvik Ray and Dr. Ruoyu Wu, Iowa State University

- Construct a diffusion approximation for a large-scale regulated matching system, characterized by a coupled regulated stochastic integral equation
- Provided a non-trivial approach to show the existence and uniqueness of the weak solution to such coupled regulated stochastic integral equations and simulated the matching behavior.
- Developing control strategies for heterogeneous matching systems, especially under resource constraints requiring admission or buffer policies.

Multi-component Matching Queue Systems in Heavy Traffic [13]

Summer 2019-2022

Advisor: Dr. Ananda Weerasinghe, Iowa State University

- Constructed many-component matching queue models and studied the diffusion-scaled heavy traffic limits, which are characterized by a coupled non-linear stochastic integral equation.
- Established the convergence of the cost functional under cost structures.
- Formulated an asymptotic relationship between the queue length processes and the virtual waiting time processes (the asymptotic Little's law) and further refined the result to L^p sense on a special probability space.
- Simulated the matching queue systems and the corresponding stochastic integral equations numerically in Python.
- Observe that a matching queue model without abandonment could possibly be interpreted as a basket option pricing problem.

INVITED TALKS

INFORMS Annual Meeting 2026

Scaling limits for regulated multi-component matching queues with abandonment

San Francisco, CA
November 1-4, 2026

Stochastic Network Conference 2026, Poster presentation

Scaling limits of regulated multi-component matching systems

Chicago Booth, IL
June 14-19, 2026

SIAM Math Club Colloquium, at The University of Akron

Inside High-Frequency Trading: The Mathematics of the Limit Order Book

Akron, OH
Mar 13, 2026

INFORMS Annual Meeting 2025

From a single-server waiting time control to reinforcement learning motivation for matching systems

Atlanta, GA
Oct 26-29, 2025

INFORMS Applied Probability Society Conference 2025

Admission control for a single server waiting time process in heavy traffic, Intro to regulated matching systems

Atlanta, GA
June 30-July 3, 2025

SIAM Math Club Colloquium, at The University of Akron

Stochastic Control in Action: Theory, Applications, and Reinforcement Learning

Akron, OH
Mar 21, 2025

Colloquium for Pure and Applied Math, at The University of Akron

Queueing Theory and Stochastic Controls: Applications in Industry, Economics, and Financial Math

Akron, OH
Oct 11, 2024

2024 Joint Mathematics Meetings, AMS Special Session

Dynamics of Matching Queues in Heavy Traffic and Some New Findings of Regulated Matching Systems

San Francisco, CA
Jan 3-6, 2024

2023 INFORMS Annual Meeting, Applied Probability Society Cluster

Multi-component matching and its dynamics and some new findings

Phoenix, AZ
Oct 15-18, 2023

Statistics and Data Science Seminar , at Washington University in St. Louis <i>Many-component Matching Queue Systems in Heavy Traffic</i>	St. Louis, MO Sep 21, 2022
SIAM Graduate Student Seminar , at Iowa State University <i>Multi-component Matching Queues in Heavy Traffic</i>	Ames, IA Apr 08, 2022
Probability and Statistics Seminar , at University of Wisconsin-Milwaukee <i>Many-component Matching Queue System</i>	Virtual Dec 1, 2021
20th Northeast Probability Seminar <i>Many-component Matching Queue System</i>	Virtual Nov 18, 2021
SIAM Graduate Student Seminar , at Iowa State University <i>Optimal Control Problem for a Diffusion Process</i>	Virtual Nov 06, 2020

COMMITTEES AND PROFESSIONAL SERVICE

INFORMS Education Outreach Graduate Committee <i>Organize webinars and in-person panel discussions at the INFORMS Annual Meeting to promote graduate education.</i>	Akron, OH Fall 2025-present
Reviewer, AMS Mathematical Reviews/MathSciNet Reviewer <i>Reviewed scholarly papers for inclusion in the AMS Mathematical Reviews database and Mathematics Subject Classification.</i>	Akron, OH Fall 2024-present
Reviewer, Methodology and Computing in Applied Probability <i>Provide peer reviews for submitted manuscripts.</i>	Akron, OH Fall 2024-present
Reviewer, SIAM Journal on Financial Mathematics <i>Provide peer reviews for submitted manuscripts.</i>	Akron, OH Fall 2024-present
Graduate Admission Committee , at The University of Akron <i>Review graduate student applications and assistantship applications in Math Department.</i>	Akron, OH Fall 2024-present
Administrative Assistant <i>Assisted the Math Department with administrative tasks, including library organization, website updates, and the Math Genealogy Project.</i>	Ames, IA Summer 2021
SIAM Student Chapter <i>Served as treasurer and participated in the planning committee for the SIAM Graduate Student Seminar.</i>	Ames, IA Fall 2020-Spring 2021

CONFERENCE ATTENDANCE

INFORMS Annual Meeting 2026, <i>San Francisco, CA</i>	November 1-4, 2026
Stochastic Network Conference 2026, <i>Chicago Booth, IL</i>	June 15-19, 2026
INFORMS Annual Meeting 2025, <i>Atlanta, GA</i>	Oct 26-29, 2025
SNAPP Summer School on Stein's Method for Queueing Approximations, <i>Virtual</i>	July 29-Aug 28, 2025
INFORMS APS Summer School, <i>Atlanta, GA</i>	June 25-29, 2025
INFORMS Applied Probability Society Conference 2025, <i>Atlanta, GA</i>	June 30-July 3, 2025
Joint Mathematics Meetings 2024, <i>San Francisco, CA</i>	Jan 2024
INFORMS Annual Meeting 2023, <i>Phoenix, AZ</i>	Oct 15-18, 2023
Seminar on Stochastic Processes, <i>Tucson, AZ</i>	Mar 8-12, 2023
First-Passage Percolation and Related Models Program, <i>ICTS, India (Hybrid)</i>	July 11-29, 2022
6th Annual Meeting of SIAM Central States Section, <i>Virtual</i>	Oct 2021

5th Annual Meeting of SIAM Central States Section, *Ames, IA*

Oct 2019

41st Midwest Probability Colloquium, *Evanston, IL*

Oct 2019

TEACHING EXPERIENCE

Teaching Experience at The University of Akron

Asst Professor, MATH 149 - Precalculus, <i>11 students</i>	Fall 2026
Asst Professor, MATH 335 - Intro to Ordinary Diff Equ, <i>44 students</i>	Fall 2026
Asst Professor, MATH 300 - Tools for Data Science, <i>16 students</i>	Spring 2026
Asst Professor, MATH 450/550 - Optimization, <i>8 undergrads, 5 grads</i>	Spring 2026
Asst Professor, MATH 335 - Intro to Ordinary Diff Equ, <i>42 students</i>	Fall 2025
Asst Professor, MATH 221 - Analytic Geometry-Calculus I, <i>34 students</i>	Fall 2025
Asst Professor, MATH 450/550 - Optimization, <i>9 students</i>	Spring 2025
Asst Professor, MATH 221 - Analytic Geometry-Calculus I, <i>40 students</i>	Spring 2025
Asst Professor, MATH 335 - Intro to Ordinary Diff Equ, <i>40 students</i>	Fall 2024
Asst Professor, MATH 221(Honors) - Analytic Geometry-Calculus I, <i>26 students</i>	Fall 2024

Teaching Experience at Washington University in St. Louis

Postdoctoral Lecturer, MATH 495/5070 - Stochastic Processes, <i>27 students</i>	Spring 2024
Postdoctoral Lecturer, MATH 493/5010 - Probability, <i>40 students</i>	Spring 2024
Postdoctoral Lecturer, MATH 5010 - Probability, <i>50 students</i>	Fall 2023
Postdoctoral Lecturer, MATH 493 - Probability, <i>8 students</i>	Summer 2023
Postdoctoral Lecturer, MATH 493 - Probability, <i>73 students</i>	Spring 2023
Postdoctoral Lecturer, MATH 3200 - Statistics and Data Analysis, <i>70 students</i>	Fall 2022
Postdoctoral Lecturer, MATH 493 - Probability, <i>48 students</i>	Fall 2022

Teaching Experience at Iowa State University

Graduate Teaching Assistant, MATH 645 - Advanced Stochastic Processes	Spring 2022
Graduate Teaching Assistant, MATH 481/581 - Numerical Methods for Diff Equ	Spring 2022
Graduate Teaching Assistant, MATH 517 - Finite Difference Methods	Spring 2022
Graduate Teaching Assistant, MATH 414 - Analysis I	Spring 2022
Graduate Teaching Assistant, MATH 554 - Intro to Stochastic Processes	Fall 2021
Graduate Teaching Assistant, MATH 561 - Numerical Analysis	Fall 2021
Recitation Instructor, MATH 143 - Preparation for Calculus	Spring 2021
Recitation Instructor, MATH 143 - Preparation for Calculus	Fall 2020
Recitation Instructor, MATH 150 - Discrete Mathematics	Spring 2020
Recitation Instructor, MATH 143 - Preparation for Calculus	Fall 2019
Lecturer, MATH 150 - Discrete Mathematics for Business and Social Sciences	Summer 2019
Recitation Instructor, MATH 151 - Calculus for Business and Social Sciences	Spring 2019
Graduate Teaching Assistant, MATH 554 - Introduction to Stochastic Processes	Fall 2018
Graduate Teaching Assistant, MATH 435 - Geometry I	Fall 2018
Graduate Teaching Assistant, MATH 373 - Introduction to Scientific Computing	Fall 2018
Graduate Teaching Assistant, MATH 104 - Introduction to Probability	Spring 2018
Graduate Teaching Assistant, MATH 304 - Combinatorics	Fall 2017
Graduate Teaching Assistant, MATH 201 - Introduction to Proofs	Fall 2017

GRADUATE COURSES

- Methods of Applied Math I, II
- Ordinary Differential Equation and Dynamic Systems
- Introduction to Stochastic Processes
- Advanced Stochastic Processes
- Machine Learning
- Numerical Analysis
- Numerical Linear Algebra
- Finite Difference Methods
- Advanced Special Topics: Continuous Time Asset Pricing Models
- Optimal Transport and Deep Learning
- Theory and Methods for Continuous Optimization (Audit)

OTHER PROJECTS

These projects are done for the laboratory parts of the Machine Learning course.

Spring 2020

- **Naive Bayes Classifier for Text Classification.** Implement the Naive Bayes algorithm to tackle the "20 Newsgroups" classification problem.
- **Decision Tree Classifier.** Experiment with the Decision Tree classifier on the congressional voting records data set using Weka.
- **Neural Network Classifier.** Using the TensorFlow package to perform an experiment on the optical recognition of the handwritten digits data set.
- **Ensemble Learning.** Train ensemble models, random forest, and AdaBoost.M1, and construct an ensemble classifier by training 4 individual models on the blood transfusion service center data set.